Evaluation Report

Simplified Lesk Word Sense Disambiguation for Closed-Caption Text — empirical results, metrics and failure analysis

1. Summary

The custom Simplified Lesk model was compared with NLTK's built-in **lesk()** on the project's annotated set (100 rows, 22 target words, 98 usable). Re-running *pipeline.py* reproduced the README figures exactly: custom **47/98 (47.96%)** versus NLTK **15/98 (15.31%)**, a gap of +32.65 percentage points (exact McNemar $p \approx 3\times10^{-6}$).

Three findings qualify that headline and are documented below:

- **Gold labels are unreliable.** Only 50 of 98 gold synset IDs match the sense that the sentence actually expresses in WordNet 3.0 (e.g. *bank.n.01* is “sloping land” but is used as gold for money-bank sentences).
- **Most predictions rest on no evidence.** 58 of 100 predictions have a zero overlap score and fall back to WordNet lemma frequency; 38 of the 47 “correct” official answers are of this kind.
- **A trivial baseline is competitive.** “Always pick the first POS-filtered WordNet sense” scores 66.33% on the official labels (above the custom model) and 50.00% on corrected labels (below it).

Conclusion: the custom model clearly beats NLTK Lesk, mainly through POS filtering, example matching and lemmatization. Its absolute accuracy is, however, not yet a trustworthy estimate until the gold labels are fixed and the test set is enlarged.

2. Experimental setup

Item	Value
Data	data/annotated.csv — 100 rows, 22 target words, 2–6 sentences each; raw_transcripts.txt (19 sentences) is not used by the pipeline
Usable rows (official)	98 — rows 60 and 62 use *goal.n.05*, which does not exist in WordNet 3.0, and are skipped by the pipeline
Software	Python 3, NLTK 3.10.3, WordNet 3.0 (the README quotes the same 47/15 results)
Custom model	Window = 5 (pipeline default; README says 7 — identical results here because sentences are short), weights w_{def}=1.0, w_{ex}=2.0, w_{syn}=0.5, POS filter on, tie-break by example overlap, definition overlap, then lemma frequency
NLTK baseline	*nltk.wsd.lesk* on the lower-cased token list: whole sentence incl. stop-words as context, definition words only (no examples, no lemmatization), no POS filter; ties go to the first WordNet sense
Metric	Accuracy = correct / evaluable rows. 95% CIs by bootstrap (5,000 resamples); paired significance by exact McNemar test

Two label sets are reported. *Official* uses the gold_synset column as supplied. *Corrected* is a supplementary audit in which the reviewer replaced each gold ID with the WordNet 3.0 synset(s) matching the intended meaning (a small accepted set where WordNet is fine-grained). It is a single-annotator judgement, not an adjudicated gold standard, and rows 63 and 95 (target word absent from the sentence as a token: “newspaper”, “classroom”) are excluded, leaving 98 rows.

3. Main results

Model	Official labels (n=98)	95% CI	Corrected labels (n=98)	95% CI
Custom Simplified Lesk	47/98 (47.96%)	37.8–58.2	57/98 (58.16%)	48.0–67.3
NLTK Lesk	15/98 (15.31%)	8.2–22.4	23/98 (23.47%)	15.3–31.6
Difference	+32.65 pp		+34.69 pp	

Table 1. Accuracy of both models under both label sets. The wide CIs (±10 pp) reflect the small test set.

Paired outcome (custom vs NLTK)	Both right	Only custom	Only NLTK	Both wrong	McNemar p
Official labels	7	40	8	43	3.3×10^{-6}
Corrected labels	15	42	8	33	1.2×10^{-6}

Table 2. The two models give the same prediction on only 21 of 100 rows. The custom model wins on the discordant rows 5:1 (official) and 5:1 (corrected).

NLTK's weakness is largely structural: it returns a synset of the wrong part of speech in 20 of 98 cases versus 1 for the custom model, and its stop-word-laden context lets words such as “the” or “of” score overlaps against definitions. Giving NLTK a POS hint raises it only to 20.41% (official) / 28.57% (corrected), so POS is not the whole story.

4. Baselines, ablations and window size

System	Official	Corrected
Custom Simplified Lesk (full)	47.96%	58.16%
First WordNet sense, POS-filtered	66.33%	50.00%
First WordNet sense, no POS filter	53.06%	42.86%
NLTK Lesk + POS hint	20.41%	28.57%
NLTK Lesk (as run in pipeline)	15.31%	23.47%
Random choice among candidates (expected)	18.3%	—

Table 3. Reference baselines. The official labels reward the most-frequent-sense prior; the corrected labels reward actual contextual evidence.

Configuration	Official	Corrected
Full model	47.96%	58.16%
No POS filter	29.59%	44.90%
No synonym term (w_s=0)	48.98%	58.16%
No example term (w_e=0)	48.98%	59.18%
No definition term (w_d=0)	47.96%	57.14%
Uniform weights (1,1,1)	47.96%	59.18%
Definition only	50.00%	59.18%
Examples only	48.98%	58.16%

Table 4. Ablation of the scoring terms. Removing the POS filter costs 18.4 pp (official) / 13.3 pp (corrected) and is the only component with a clear effect. Changes from the definition, example and synonym terms and their weights are at most 2 pp (≤ 2 rows) — within noise — because the context rarely contains more than one or two overlapping words.

Window	1	2	3	5	7	10	15
Official %	67.3	61.2	55.1	48.0	48.0	48.0	48.0
Corrected %	53.1	53.1	56.1	58.2	58.2	58.2	58.2

Table 5. Context-window sweep. Accuracy is constant from window 5 upward (sentences have ≤ 11 tokens). On official labels a smaller window increases accuracy, because fewer overlaps means more frequency-fallback answers, which the labels favour; corrected labels show the expected opposite trend.

5. Per-word results

Word	Official custom	Official NLTK	Corrected custom	Corrected NLTK
address	1/2	0/2	0/2	0/2
arm	2/4	0/4	3/4	0/4
ball	2/4	2/4	2/4	2/4
bank	3/5	3/5	3/5	3/5
bark	3/5	0/5	4/5	0/5
bass	1/4	0/4	2/4	1/4
bat	1/5	2/5	2/5	0/5
class	1/4	0/4	2/3	2/3
crane	4/5	0/5	3/5	1/5
field	1/4	0/4	1/4	0/4
goal	0/2	0/2	4/4	3/4
hoe	4/4	2/4	4/4	2/4
light	3/5	0/5	3/5	0/5
match	2/5	0/5	3/5	0/5
note	2/4	1/4	2/4	2/4
paper	1/4	1/4	0/3	1/3

Word	Official custom	Official NLTK	Corrected custom	Corrected NLTK
plant	3/6	2/6	3/6	2/6
rock	3/6	1/6	4/6	1/6
spring	2/6	1/6	4/6	2/6
table	2/4	0/4	2/4	1/4
track	2/4	0/4	1/4	0/4
watch	4/6	0/6	5/6	0/6

Table 6. Correct/total per target word. Strong: hoe (4/4 both label sets), watch, bark, rock, spring (corrected). Weak: paper (0/3), address (0/2), field (1/4), track (1/4). Per-word counts are 2–6 rows, so single words are anecdotal.

6. Gold-label audit

Of the 98 usable rows, the gold ID agrees with the intended sense in **50** and disagrees in 48. The errors are systematic: the file groups sentences by intended meaning but attaches the wrong sense number (sense IDs appear to have been assigned from memory rather than checked against WordNet 3.0). Effect on the official score: 22 rows marked wrong are actually right, and 14 marked right are actually wrong (including rows 63 and 95, which have no target token yet are counted correct); 33 are right and 29 wrong under both. The net effect is small (47 → 57 correct), but it hides large per-word distortions.

Word	Gold ID used	What that ID means in WordNet 3.0	Rows	Intended sense
address	address.n.01	(computer science) the code that identifies wh…	100	address.n.02, address.n.06
address	address.v.01	speak to	99	address.v.05
arm	arm.n.02	any projection that is thought to resemble a h…	73, 74	branch.n.01
bank	bank.n.01	sloping land (especially the slope beside a bo…	1, 4, 5	depository_financial_institution.n.01
bark	bark.v.01	speak in an unfriendly tone	26, 27, 28	bark.v.04
bass	bass.n.01	the lowest part of the musical range	91, 94	bass.n.08, freshwater_bass.n.01
bass	bass.n.02	the lowest part in polyphonic music	92, 93	bass.n.01, bass.n.07
bat	bat.n.01	nocturnal mouselike mammal with forelimbs modi…	9, 10	bat.n.05
bat	bat.n.02	(baseball) a turn trying to get a hit	6, 7, 8	bat.n.01
class	class.n.01	a collection of things sharing a common attrib…	95, 97	class.n.02, course.n.01
class	class.n.03	people having the same social, economic, or ed…	98	class.n.01
crane	crane.n.01	United States writer (1871-1900)	24, 25	crane.n.05
field	field.n.05	the space around a radiating body within which…	80, 82	discipline.n.01, sphere.n.01
goal	goal.n.01	the state of affairs that a plan is intended t…	59, 61	goal.n.03, goal.n.04
match	match.n.01	lighter consisting of a thin piece of wood or …	18, 19, 20	match.n.02
match	match.n.02	a formal contest in which two or more persons …	16, 17	match.n.01
paper	paper.n.02	an essay (especially one written as an assignm…	65	paper.n.05
rock	rock.n.02	material consisting of the aggregate of minera…	40, 41, 42	rock_'n'_roll.n.01
spring	spring.n.03	a natural flow of ground water	43, 44, 45	spring.n.02
table	table.n.01	a set of data arranged in rows and columns	75, 77	table.n.02
table	table.n.02	a piece of furniture having a smooth flat top …	76, 78	table.n.01
track	track.n.01	a line or route along which something travels …	85	track.n.03, track.n.09

Word	Gold ID used	What that ID means in WordNet 3.0	Rows	Intended sense
track	track.v.01	carry on the feet and deposit	84, 86	chase.v.01, track.v.02

Table 7. Gold IDs whose WordNet meaning conflicts with the sentences they label (23 groups, 47 rows). Remaining rows (51) agree.

7. Failure analysis

On the corrected labels the custom model fails on 41 rows plus the 2 target-absent rows. Failures fall into four categories:

Category	Rows	Typical cause / example
C. Zero overlap → frequency fallback chose wrong sense	23	No context word appears in any sense signature, so lemma counts decide; the most frequent sense is wrong. E.g. “The bank approved his loan application” → *bank.n.01* (slope; count 25 vs 20 for the financial sense); “The plant needs water…”, “wooden table”, “charity ball”, “football match”.
D. Spurious or insufficient overlap favoured wrong sense	17	A generic word overlaps with a wrong sense: “hit” → *bat.n.02* (baseball turn) for a sports bat (score 3.0, highest in the set); “sheet”/“read” → *newspaper.n.01* for paper; “near” → *grus.n.01* for the crane bird; “high” → *eminence.n.01* for a musical note; “please”, “form”, “work”, “carry”, “run”.
A. Target token absent	2	Rows 63 (“newspaper”) and 95 (“classroom”): the whole-token match fails, context is empty, and the model returns the first sense.
B. Correct sense filtered out	1	Row 41: “listening to rock music” — “rock” is tagged as a verb, so only verb senses are considered. Row 68 (“Please note…”) received no POS and fell back to all senses.

Table 8. Failure taxonomy (corrected labels).

Observations

- **The frequency fallback dominates.** With a mean of 3.1 context words against about 7.7 candidate senses per word, WordNet glosses seldom share vocabulary with the context. Accuracy with zero overlap is 57.1% (corrected), and higher-scoring buckets are not reliably better: 50.0% for scores 0.5–1, 72.2% for 1.5–2.5, 50.0% for ≥ 3 (official: 66.7%, 21.1%, 27.8%, 0.0%). The score is therefore a poor confidence signal.
- **Exact-match lexical overlap is brittle.** Related words (fish/lake for bass, guitar/band for music, wings for bird) do not match gloss vocabulary; function-like words (“please”, “near”, “work”) match by chance.
- **Target-word leakage is only partial.** Only the first occurrence of the target token is removed. In “She planted a small plant…” the lemma of “planted” is “plant”, which matches the examples of *plant.n.01* (score 2.5) and causes an error.
- **Lemmatization asymmetry.** Context tokens are lemmatized with their own tagged POS, but gloss and example tokens are lemmatized with the target word's POS, so forms can fail to align.
- **NLTK failures** are mostly cross-POS errors (20 of 98), e.g. verb senses for “bat” or “bank”, and stop-word overlaps that favour long glosses; it never uses example sentences.
- **Not implemented.** The README describes phrase matching and a WordNet-similarity bonus; neither exists in models.py, so they are not evaluated here.

8. Threats to validity

- Small sample (98 rows, 22 words, 2–6 sentences each); 95% intervals span about 20 pp.
- Official gold labels are mostly wrong IDs; the corrected set is a single-reviewer judgement and may be debatable for fine-grained senses (accepted sets were used where WordNet splits meanings finely).
- Test sentences were written to be unambiguous and short, unlike noisy caption text; results will not transfer directly to real captions.
- Hyper-parameters were not tuned on a separate split; only the ablation/window sweeps above were run.

9. Recommendations

- Correct the gold IDs (Table 7), remove or fix rows 60, 62, 63 and 95, and report accuracy on the corrected set alongside the first-sense baseline.
- Replace zero-score fallback by an explicit “low confidence” flag; report coverage vs accuracy.

- Add stop-word-like filtering of generic words and match whole-sentence leakage by lemma; fix POS fallback for tags that do not map (e.g. interjections).
- Extend the signature with hypernym/hyponym glosses (extended Lesk) or embedding similarity to fix lexical gaps; evaluate on SemCor / Senseval / SemEval for credible numbers.
- Align README with the code (window size, file names, implemented features) and delete src/tempCodeRunnerFile.py (a duplicate of models.py).

10. Reproduction

`cd src && python pipeline.py` prints the summary (custom 47/98, NLTK 15/98) and writes data/annotated/evaluation_results.csv. Ablation, window, baseline and audit numbers in this report were produced with the same models.py/preprocessing.py by varying constructor arguments; the corrected labels are listed in Table 7 and the failure rows in Table 8.